## AD 2. AERODROMES

### VEGT AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VEGT - LOKPRIYA GOPINATH BORDOLOI INTERNATIONAL AIRPORT, GUWAHATI

### VEGT AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

| | | | |
|---|---|---|---|
| 1 | Aerodrome reference point coordinates and its site | 260618N 0913508E<br>MAG. BRG. 27.5 DEG /1269M From Physical Beginning of RWY 02 | |
| 2 | Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves | 242 DEG,18Km from Guwahati Railway station | |
| 3 | Aerodrome elevation and reference temperature | 163 FT / 34.5 DEG C | |
| 4 | Magnetic variation, date of information and annual change | 0.36 DEG W (2010) /0.00 | |
| 5 | Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available) | Guwahati International Airport Ltd. Chief Airport Officer<br>New MT Pool Building,<br>Lokpriya Gopinath Bordoloi International Airport<br>Guwahati -781015, | |
| | | Telephone: | (91) 3612840009 |
| | | Fax: | (91) 3612840009 |
| | | AFS: | VEGTYHYX. |
| | | Email: | cao.guwahatiairport@adani.com |
| 6 | Types of traffic permitted (IFR/VFR) | IFR/VFR | |
| 7 | Remarks | 1)Vertical Reference System: EGM-08.<br>2) Aerodrome Reference Code: 4D. | |

### VEGT AD 2.3 OPERATIONAL HOURS

| | | |
|---|---|---|
| 1 | Aerodrome Operator | MON-FRI 0400-1300 UTC (0930-1830 IST)<br>SAT 0400-0800 UTC (0930-1330 IST)<br>SUN/HOLIDAY: NIL |
| 2 | Custom and immigration | H24, See remark no. 1 |
| 3 | Health and sanitation | H24 |
| 4 | AIS briefing office | H24 |
| 5 | ATS reporting office (ARO) | H24 |
| 6 | MET Briefing office | H24 |
| 7 | Air Traffic Service | H24 |
| 8 | Fuelling | H24 |
| 9 | Handling | H24 |
| 10 | Security | H24 |
| 11 | De-icing | NIL |

| | | |
|---|---|---|
| 12 | Remarks | 1. Custom and Immigration facilities are provided on limited basis to cover operations of schedule flights. The facilities can be arranged to cover any authorized non-sked operations with prior notice to AD<br><br>2.Non- sked flights to obtain positive clearance 24 hr before operating to Guwahati.<br>3. Approved Hourly RWY Traffic Handing Capacity for W25<br>Runway 02:<br>Maximum no. of Arrival & Departure : 19<br>Maximum no. of Arrivals only : 11<br>Maximum no. of Departures only : 13<br>Runway 20:<br>Maximum no. of Arrival & Departure : 15<br>Maximum no. of Arrivals only : 08<br>Maximum no. of Departures only : 10<br>3. AIASL (Air India Airport Services limited).<br>Indo Thai Airport Management services Pvt Ltd.<br>4. On T2 Apron, EV tugs, Fuel hydrant, Passenger boarding bridges, A-VDGS are available. |

## VEGT AD 2.4 HANDLING SERVICES AND FACILITIES

| | | |
|---|---|---|
| 1 | Cargo-handling facilities | AAICLAS CUDCT and International Cargo<br>Cargo Terminal Area 8400 sq m, Capacity to handle approx. 39200 MT cargo yearly.<br>GIAL Domestic Cargo<br>Cargo Terminal Area 628 sq m, Capacity to handle approx. 2750 MT cargo yearly.<br><br>Indo Thai Airport Management services Pvt Ltd. |
| 2 | Fuel and Oil types | JET A1 |
| 3 | Fuelling facilities and capacity | Adani Aviation Fuel Services Limited<br>Two Aviation Fuelling Stations available with a Total Capacity of 1020.5 KL.<br>Refuelers / Bowsers of various capacity ranging from 6 KL to 25 KL available. |
| 4 | De-icing facilities | NIL |
| 5 | Hangar space for visiting aircraft | Nil |
| 6 | Repair facilities for visiting aircraft | Nil |
| 7 | Remarks | NIL |

## VEGT AD 2.5 PASSENGER FACILITIES

| | | |
|---|---|---|
| 1 | Hotel(s) at or in the vicinity of aerodrome | Available near the Airport and In the city. |
| 2 | Restaurant(s) at or in the vicinity of aerodrome | At AD and in the city. |
| 3 | Transportation possibilities | Taxis and ASTC buses. |
| 4 | Medical Facilities | First aid at AD. Hospitals in the city. |
| 5 | Bank and post office at or in the vicinity of aerodrome | Banks: Near AD & in City<br>Post office: Near AD & in City |
| 6 | Tourist office | At AD. |
| 7 | Remarks | NIL |

## VEGT AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

| | | |
|---|---|---|
| 1 | Aerodrome category for fire fighting | Within ATS HR: CAT-7 |

| | | |
|---|---|---|
| 2 | Rescue equipment | Available as per category. |
| 3 | Capability for removal of disabled aircraft | 1. The critical aircraft for the airport is A321-200 NEO.<br>2. Primary responsibility for removal of disabled aircraft rests with the concerned airline.<br>3. The airlines may contact 'Aerodrome Coordinator for Disabled Aircraft Removal Operations' for coordinating with the agencies having Disabled Aircraft Recovery Kit.<br>4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations:<br>[Aerodrome Coordinator:Mr. Sanjeev Kumar Gupta,<br>Head-Operations,<br>Guwahati International Airport Limited.<br>Contact details.:+91 9029266499<br>Email:sanjeev.gupta@adani.com, arff.gau@adani.com |
| 4 | Remarks | NIL |

## VEGT AD 2.7 SEASONAL AVAILABILITY CLEARING

| | | |
|---|---|---|
| 1 | Type(s) of clearing equipment | Manual sweeping and Runway Sweeper Machine |
| 2 | Clearance priorities | Available 24x7 |
| 3 | Remarks | NIL |

## VEGT AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

| | | |
|---|---|---|
| 1 | Designation, surface and strength of aprons | Refer Aircraft Parking/Docking Chart |
| 2 | Designation, width, surface and strength of taxiways | Refer Aircraft Parking/Docking Chart |
| 3 | Location and elevation of altimeter checkpoints | Location: All Stands<br>Elevation: Refer parking and docking charts of Aprons |
| 4 | Location of VOR checkpoints | Available on TWY E, F, G, H and J |
| 5 | Position of INS checkpoints | |
| 6 | Remarks | 1. Coordinates of RWY Holding Position:<br>i. TWY E: 260626.156N 0913511.417E/ELEV: 161FT<br>ii. TWY F: 260620.367N 0913508.752E /ELEV: 161FT<br>iii. TWY G: 260711.901N 0913533.891E/ELEV: 162FT<br>2. Coordinates of RWY & Taxi Intersection:<br>i. TWY E: 260627.291N 0913508.422E/ELEV: 163FT<br>ii. TWY F: 260621.498N 0913505.740E /ELEV: 163FT<br>iii. TWY G: 260713.414N 0913529.821E/ELEV: 163FT<br><br>3. Gravity Model: EGM08.<br>4. All Taxiways (L1, L2, L3 & L4) on the west side of Airfield are for Military use Only. |

## VEGT AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

| | | |
|---|---|---|
| 1 | Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands | i) Illuminated stand identification boards displaying stand number and stand co-ordinates are available for aircraft stands 1,2,3,4,5,6,7,8 and 9, on Apron I, and Stands 21L, 21, 21R, 22L, 22R, 23L, 27 and 28 on T2 Apron. Stand identification marking available on ground for Stands 41, 42, 43, 44, and 45 on Apron K<br>ii) Taxiing guidance signs available at all intersection of TWY and RWY and at holding positions.<br>iii) Advanced Visual Docking Guidance system (A-VDGS) installed on stands 2 on Apron I, and stands 21L, 21, 21R, 22L, 22R, 23L, 27 & 28 on T2 Apron.<br>iv) Marshaling guidance available for Stand 1, 6, 7, 8 and 9 on Apron I, and Stand 41, 42, 43, 44, and 45 through authorized Ground Handing agents<br>(GHA).<br>v) Aircraft Stand Lead-In Lines available at apron. |
| 2 | Runway and taxiway markings and lights | RWY<br><br>Markings: Designation, THR,Transverse stripe, TDZ, Centre line,Side Stripe, Aiming Point.<br>Lights: THR, Edge, End<br><br>TWY<br><br>Markings: Center line, RWY Holding Positions Intermediate Holding Position, Edge (Enhanced RWY Holding Positions are marked on all TWYs connected with RWYs)<br>Lights: Edge lights available on E, F, G, H, J, B9, C8, portion of Taxiway A. |
| 3 | Stop bars (if any) | Nil |
| 4 | Remarks | A-VDGS / VDGS is operational H24 in all-weather / visibility conditions. NOTAM is promulgated whenever A-VDGS / VDGS is unserviceable. |

## VEGT AD 2.10 AERODROME OBSTACLES

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| 20/TKOF<br>02/APCH | TREE | 255904.9N<br>0913230.7E | 1271 FT | NIL | TREE ON HILL |
| 20/TKOF<br>02/APCH | TREE | 255758.2N<br>0913236.7E | 1264 FT | NIL | TREE ON HILL |
| 20/TKOF<br>02/APCH | OTHER | 260535.6N<br>0913443.5E | 178 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV - 49.16 M + 5 M MOBILE TRAFFIC) |
| 20/TKOF<br>02/APCH | OTHER | 260535.8N<br>0913443.2E | 178 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV - 49.16 M + 5 M MOBILE TRAFFIC) |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| 20/TKOF<br>02/APCH | OTHER | 260536.0N<br>0913442.7E | 178 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV - 49.13 M + 5 M MOBILE TRAFFIC) |
| 20/TKOF<br>02/APCH | OTHER | 260536.1N<br>0913442.2E | 178 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV - 49.17 M + 5 M MOBILE TRAFFIC) |
| 20/TKOF<br>02/APCH | OTHER | 260534.1N<br>0913447.6E | 177 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV - 49.03 M + 5 M MOBILE TRAFFIC) |
| 20/TKOF<br>02/APCH | OTHER | 260535.3N<br>0913446.5E | 175 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV - 48.28 M + 5 M MOBILE TRAFFIC) |
| 20/TKOF<br>02/APCH | OTHER | 260535.5N<br>0913445.9E | 175 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV - 48.25 M + 5 M MOBILE TRAFFIC) |
| 20/TKOF<br>02/APCH | OTHER | 260535.7N<br>0913445.3E | 175 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV - 48.25 M + 5 M MOBILE TRAFFIC) |
| 20/TKOF<br>02/APCH | OTHER | 260536.0N<br>0913444.6E | 175 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV - 48.25 M + 5 M MOBILE TRAFFIC) |
| 20/TKOF<br>02/APCH | OTHER | 260536.1N<br>0913444.1E | 175 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV - 48.26 M + 5 M MOBILE TRAFFIC) |
| 20/TKOF<br>02/APCH | FENCE | 260539.4N<br>0913442.4E | 167 FT | NIL | AIRPORT BOUNDARY FENCE |
| 20/TKOF<br>02/APCH | OTHER | 260534.9N<br>0913447.7E | 175 FT | NIL | MOBILE ROAD TRAFFIC(ROAD ELEV - 48.25 M + 5M MOBILE TRAFFIC) |
| 20/TKOF<br>02/APCH | OTHER | 260537.2N<br>0913441.9E | 175 FT | NIL | MOBILE ROAD TRAFFIC(ROAD ELEV - 48.27 M + 5M MOBILE TRAFFIC) |
| 20/APCH<br>02/TKOF | OTHER | 260718.4N<br>0913529.5E | 177 FT | NIL | MOBILE ROAD TRAFFIC(ROAD ELEV - 49.03 M + 5M MOBILE TRAFFIC) |
| 20/APCH<br>02/TKOF | OTHER | 260716.9N<br>0913528.8E | 178 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV. 48.9 M. +5M) |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | BUILDING | 260542.7N 0913636.7E | 323 FT | LGTD | SHINE SANSAAR, SINTEX TANK TOP - 98.35 M, STAIRCASE ROOM TOP - 96.32 M, DESIGN TOP - 94.54 M, BUILDING TOP - 93.15 M. |
| In circling area and at AD | BUILDING | 260541.4N 0913636.3E | 322 FT | LGTD | SHINE SANSAAR, SINTEX TANK TOP – 98.09 M, STAIRCASE ROOM TOP – 96.38 M, DESIGN TOP – 94.37 M, SINTEX TANK TOP – 93.67 M, BUILDING TOP – 92.80 M. |
| In circling area and at AD | BUILDING | 261005.6N 0914215.6E | 883 FT | NIL | BUILDING ON HILL |
| In circling area and at AD | BUILDING | 260702.1N 0913519.6E | 172 FT | NIL | BUILDING, SINTEX TANK TOP – 52.32 M, SINTEX TANK TOP – 52.24 M, LIGHT POLE – 52.14 M, LIGHT POLE – 52.20 M, LIGHT POLE – 52.19 M, LIGHT POLE – 52.04 M, BUILDING TOP – 51.37 M. |
| In circling area and at AD | BUILDING | 260545.7N 0913444.3E | 178 FT | NIL | BUILDING, LADDER TOP - 54.32 M, SINTEX TANK TOP - 52.50 M, BUILDING TOP - 51.58 M. |
| In circling area and at AD | OTHER | 260445.0N 0913629.7E | 343 FT | NIL | CELLPHONE MAST (AIRTEL) |
| In circling area and at AD | OTHER | 260943.8N 0914331.0E | 1005 FT | NIL | CELLPHONE MAST ON HILL |
| In circling area and at AD | OTHER | 261138.6N 0914626.5E | 967 FT | NIL | CELLPHONE MAST ON HILL |
| In circling area and at AD | OTHER | 261428.3N 0913222.8E | 840 FT | NIL | CELLPHONE MAST ON HILL (AIRTEL) |
| In circling area and at AD | OTHER | 261431.6N 0913224.4E | 868 FT | NIL | CELLPHONE MAST ON HILL (BSNL) |
| In circling area and at AD | TREE | 260121.4N 0913447.5E | 1335 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260121.6N 0913505.6E | 1112 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260110.1N 0913511.1E | 1214 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260111.0N 0913525.9E | 937 FT | NIL | TREE ON HILL |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | TREE | 260026.6N 0913541.3E | 1739 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260028.6N 0913552.3E | 1609 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 255953.0N 0913636.9E | 1139 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260101.4N 0913621.9E | 1653 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260100.6N 0913642.6E | 1557 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260106.8N 0913656.1E | 1471 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260152.3N 0913556.9E | 755 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260204.8N 0913613.4E | 899 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260219.7N 0913616.9E | 1078 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260236.9N 0913556.5E | 1018 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260235.7N 0913533.2E | 620 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260245.6N 0913534.4E | 630 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260243.6N 0913614.0E | 1065 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260243.7N 0913631.6E | 1084 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260243.6N 0913649.2E | 1229 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260242.6N 0913700.3E | 1164 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260245.5N 0913705.5E | 1168 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260307.7N 0913627.2E | 829 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260324.7N 0913618.9E | 509 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260334.7N 0913656.1E | 688 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260332.4N 0913652.4E | 648 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260429.9N 0913640.7E | 712 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260526.6N 0913709.0E | 475 FT | NIL | TREE ON HILL |
| In circling area and at AD | OTHER | 260437.9N 0913631.6E | 341 FT | NIL | DURGA MANDIR ON HILL |
| In circling area and at AD | TREE | 260556.4N 0913725.2E | 688 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260542.9N 0913718.8E | 545 FT | NIL | TREE ON HILL |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | TREE | 260433.2N 0913649.0E | 728 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260442.0N 0913710.4E | 675 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260449.6N 0913726.4E | 732 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260459.5N 0913711.0E | 555 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260509.4N 0913729.7E | 565 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260517.9N 0913741.7E | 600 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260540.7N 0913812.1E | 692 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260543.1N 0913753.8E | 694 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260535.1N 0913730.0E | 567 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261135.8N 0913907.2E | 758 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261136.0N 0913914.7E | 746 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261137.4N 0913918.8E | 766 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261157.0N 0913921.5E | 728 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261207.9N 0913924.6E | 740 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261212.6N 0913934.1E | 657 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261208.6N 0913941.0E | 698 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261229.4N 0913959.2E | 915 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261232.7N 0914007.9E | 1014 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261308.4N 0914026.4E | 1277 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261321.0N 0914045.3E | 1204 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261313.0N 0914120.3E | 869 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261309.6N 0914127.5E | 1100 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260946.4N 0914224.4E | 723 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261011.6N 0914150.3E | 674 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261003.0N 0914202.8E | 678 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260903.8N 0914328.2E | 931 FT | NIL | TREE ON HILL |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | TREE | 260913.3N 0914335.4E | 770 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260922.0N 0914240.8E | 673 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260617.9N 0913852.6E | 774 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260609.7N 0913858.3E | 817 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260552.7N 0913903.0E | 1056 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260556.2N 0913915.9E | 995 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260555.0N 0913937.1E | 939 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260816.3N 0914120.7E | 748 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260817.1N 0914236.0E | 855 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260822.7N 0914243.6E | 868 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260833.6N 0914246.5E | 862 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260838.5N 0914246.6E | 778 FT | NO | TREE ON HILL |
| In circling area and at AD | TREE | 260841.9N 0914312.7E | 981 FT | NIL | TREE ON HILL |
| In circling area and at AD | OTHER | 260727.2N 0914258.9E | 663 FT | NIL | HILL TOP |
| In circling area and at AD | TREE | 260548.7N 0913802.2E | 568 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260543.5N 0913839.9E | 662 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260605.5N 0913917.6E | 880 FT | NIL | Tree on Hill |
| In circling area and at AD | TREE | 260544.6N 0914300.8E | 830 FT | NIL | Tree on Hill |
| In circling area and at AD | TREE | 260526.5N 0914216.4E | 1245 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260539.7N 0914216.2E | 764 FT | NIL | Tree on Hill |
| In circling area and at AD | TREE | 260523.4N 0914204.2E | 1256 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260525.8N 0914133.6E | 1297 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260518.3N 0914147.5E | 1179 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260528.6N 0914300.8E | 1101 FT | NIL | Tree on Hill |
| In circling area and at AD | TREE | 255651.5N 0913522.6E | 2314 FT | NIL | Tree on Hill |
| In circling area and at AD | TREE | 260039.8N 0913437.3E | 1057 FT | NIL | TREE ON HILL |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | TREE | 260014.8N 0913431.0E | 1163 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260017.0N 0913404.9E | 915 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 255809.3N 0913313.5E | 1312 FT | NIL | Tree on Hill |
| In circling area and at AD | TREE | 255829.0N 0913539.1E | 1915 FT | NIL | Tree on Hill |
| In circling area and at AD | TREE | 255803.2N 0913116.0E | 1348 FT | NIL | Tree on Hill |
| In circling area and at AD | TREE | 255631.7N 0912539.8E | 1342 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 255835.2N 0912703.4E | 865 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 255844.0N 0912706.3E | 915 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 255657.2N 0913116.1E | 1763 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260419.7N 0913213.3E | 473 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260439.1N 0913223.2E | 743 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261431.7N 0913223.6E | 780 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261135.7N 0913901.6E | 756 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 261030.8N 0912849.6E | 780 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260438.3N 0913223.7E | 734 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260500.3N 0913236.8E | 613 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260515.8N 0913232.2E | 779 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260445.6N 0913316.6E | 414 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260457.0N 0913221.3E | 660 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260247.4N 0914111.4E | 1900 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260250.8N 0914129.4E | 1704 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260431.0N 0913807.5E | 729 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260436.2N 0913808.1E | 689 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260449.2N 0914124.3E | 1381 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260238.9N 0914221.9E | 2116 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260354.0N 0914244.3E | 1358 FT | NIL | TREE ON HILL |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | TREE | 260301.0N 0913811.2E | 1081 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260320.1N 0913640.7E | 881 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260322.2N 0914033.7E | 1480 FT | NIL | TREE ON HILL |
| In circling area and at AD | TREE | 260317.1N 0914056.9E | 1600 FT | NIL | TREE ON HILL |
| In circling area and at AD | SIGN | 260713.1N 0913534.4E | 164 FT | NIL | SIGN BOARD |
| In circling area and at AD | SIGN | 260711.0N 0913533.3E | 165 FT | NIL | SIGN BOARD |
| In circling area and at AD | SIGN | 260710.8N 0913533.3E | 164 FT | NIL | SIGN BOARD |
| In circling area and at AD | SIGN | 260625.4N 0913511.0E | 163 FT | NIL | SIGN BOARD |
| In circling area and at AD | SIGN | 260627.0N 0913511.8E | 164 FT | NO | SIGN BOARD |
| In circling area and at AD | SIGN | 260627.0N 0913511.9E | 163 FT | NIL | SIGN BOARD |
| In circling area and at AD | SIGN | 260618.6N 0913505.8E | 164 FT | NIL | SIGN BOARD |
| In circling area and at AD | SIGN | 260623.5N 0913508.0E | 164 FT | NIL | SIGN BOARD |
| In circling area and at AD | SIGN | 260624.4N 0913508.5E | 164 FT | NIL | SIGN BOARD |
| In circling area and at AD | SIGN | 260629.3N 0913510.7E | 165 FT | NIL | SIGN BOARD |
| In circling area and at AD | OTHER | 260704.2N 0913527.0E | 163 FT | NIL | PAPI |
| In circling area and at AD | OTHER | 260704.1N 0913527.3E | 163 FT | NIL | PAPI |
| In circling area and at AD | OTHER | 260704.0N 0913527.6E | 163 FT | NIL | PAPI |
| In circling area and at AD | OTHER | 260703.9N 0913527.9E | 163 FT | NIL | PAPI |
| In circling area and at AD | SIGN | 260619.5N 0913508.4E | 163 FT | NIL | Sign Board |
| In circling area and at AD | SIGN | 260619.4N 0913508.4E | 163 FT | NIL | Sign Board |
| In circling area and at AD | SIGN | 260703.5N 0913521.9E | 163 FT | NIL | Sign Board |
| In circling area and at AD | SIGN | 260704.4N 0913524.2E | 164 FT | NIL | Sign Board |
| In circling area and at AD | SIGN | 260641.7N 0913513.6E | 164 FT | NIL | Sign Board |
| In circling area and at AD | SIGN | 260605.0N 0913454.6E | 163 FT | NIL | Sign Board |
| In circling area and at AD | SIGN | 260603.6N 0913453.9E | 163 FT | NIL | Sign Board |
| In circling area and at AD | SIGN | 260602.9N 0913455.6E | 164 FT | NIL | Sign Board |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | SIGN | 260604.4N 0913456.3E | 163 FT | NIL | Sign Board |
| In circling area and at AD | SIGN | 260621.2N 0913509.1E | 164 FT | NIL | Sign Board |
| In circling area and at AD | SIGN | 260621.2N 0913509.1E | 164 FT | NIL | Sign Board |
| In circling area and at AD | OTHER | 260550.1N 0913449.7E | 163 FT | NIL | PAPI |
| In circling area and at AD | OTHER | 260550.2N 0913449.4E | 163 FT | NIL | PAPI |
| In circling area and at AD | OTHER | 260550.3N 0913449.1E | 163 FT | NIL | PAPI |
| In circling area and at AD | OTHER | 260550.5N 0913448.8E | 163 FT | NIL | PAPI |
| In circling area and at AD | OTHER | 260623.0N 0913509.8E | 163 FT | NIL | LANDING T TOP |
| In circling area and at AD | POLE | 260709.9N 0913533.1E | 171 FT | NIL | CAMERA POLE |
| In circling area and at AD | SIGN | 260709.8N 0913533.5E | 167 FT | NIL | SIGN BOARD |
| In circling area and at AD | OTHER | 260710.9N 0913532.1E | 164 FT | NIL | ELECTRICAL BOX |
| In circling area and at AD | FENCE | 260710.2N 0913532.0E | 168 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | FENCE | 260706.2N 0913530.1E | 169 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | FENCE | 260709.1N 0913531.5E | 168 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | SIGN | 260708.7N 0913530.9E | 166 FT | NIL | Sign Board |
| In circling area and at AD | OTHER | 260651.5N 0913522.5E | 167 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | SIGN | 260646.2N 0913520.9E | 163 FT | NIL | Sign Board |
| In circling area and at AD | FENCE | 260647.2N 0913521.5E | 168 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | SIGN | 260639.6N 0913517.6E | 164 FT | NIL | Sign Board |
| In circling area and at AD | SIGN | 260638.5N 0913519.0E | 167 FT | NIL | ELECTRICAL BOX |
| In circling area and at AD | OTHER | 260638.5N 0913518.9E | 166 FT | NIL | ELECTRICAL BOX |
| In circling area and at AD | SIGN | 260636.4N 0913517.5E | 163 FT | NIL | Sign Board |
| In circling area and at AD | POLE | 260635.5N 0913517.6E | 173 FT | NIL | POLE TOP |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | POLE | 260644.2N 0913520.7E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260643.6N 0913520.6E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | OTHER | 260642.9N 0913520.4E | 164 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260642.3N 0913520.2E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260641.7N 0913520.0E | 164 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260641.1N 0913519.7E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260640.5N 0913519.5E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260639.9N 0913519.3E | 164 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260639.1N 0913519.0E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260638.5N 0913518.7E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260637.9N 0913518.5E | 164 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260636.7N 0913518.0E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260636.1N 0913517.8E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260635.5N 0913517.5E | 164 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260634.8N 0913517.3E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260634.2N 0913517.0E | 164 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260633.6N 0913516.8E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260639.0N 0913519.0E | 165 FT | NIL | POLE TOP |
| In circling area and at AD | POLE | 260629.7N 0913514.8E | 167 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | POLE | 260646.8N 0913521.2E | 164 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260646.1N 0913521.1E | 164 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260645.5N 0913521.0E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260644.8N 0913520.8E | 165 FT | NIL | LIGHT POLE |
| In circling area and at AD | OTHER | 260612.9N 0913506.0E | 167 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | OTHER | 260615.8N 0913507.0E | 167 FT | NIL | HUT |
| In circling area and at AD | SIGN | 260616.0N 0913506.7E | 166 FT | NIL | SIGN BOARD |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | POLE | 260616.4N 0913506.9E | 164 FT | NIL | SIGNAL POLE |
| In circling area and at AD | OTHER | 260549.3N 0913455.9E | 166 FT | NIL | ELECTRICAL JUNCTION BOX |
| In circling area and at AD | OTHER | 260551.9N 0913455.9E | 167 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | POLE | 260554.3N 0913457.3E | 171 FT | NIL | LIGHTNING ARRESTER |
| In circling area and at AD | ANTENNA | 260554.8N 0913457.4E | 184 FT | LGTD | MET ANTENNA, LIGHTNING ARRESTER TOP - 56.19 M, OBSTRUCTION LIGHT TOP - 54.52 M, MET ANTENNA TOP - 54.19 M. |
| In circling area and at AD | OTHER | 260554.8N 0913457.2E | 168 FT | NIL | TRANSMISSOMETER |
| In circling area and at AD | OTHER | 260553.9N 0913456.8E | 168 FT | NIL | TRANSMISSOMETER |
| In circling area and at AD | OTHER | 260554.9N 0913457.2E | 165 FT | NIL | ELECTRICAL JUNCTION BOX |
| In circling area and at AD | OTHER | 260555.1N 0913457.3E | 165 FT | NIL | ELECTRICAL JUNCTION BOX |
| In circling area and at AD | FENCE | 260554.1N 0913458.5E | 171 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | POLE | 260540.3N 0913450.6E | 166 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | FENCE | 260539.5N 0913451.5E | 163 FT | NIL | AIRPORT BOUNDARY WALL |
| In circling area and at AD | FENCE | 260539.8N 0913450.5E | 163 FT | NIL | AIRPORT BOUNDARY WALL |
| In circling area and at AD | ANTENNA | 260712.7N 0913525.6E | 178 FT | NIL | ANTENNA, LIGHTNING ARRESTER TOP - 54.31 M, ANTENNA TOP - 54.09 M. |
| In circling area and at AD | POLE | 260712.6N 0913525.9E | 173 FT | NIL | DGPS ANTENNA TOP |
| In circling area and at AD | POLE | 260712.4N 0913525.5E | 177 FT | NIL | POLE TOP |
| In circling area and at AD | BUILDING | 260712.5N 0913525.7E | 172 FT | NIL | INRES SITE, SHED TOP |
| In circling area and at AD | POLE | 260712.1N 0913525.6E | 182 FT | LGTD | ASMGCS (RU - 06), LIGHTNING ARRESTER TOP - 55.41 M, OBSTRUCTION LIGHT TOP - 55.10 M, ANTENNA TOP - 54.93 M. |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | BUILDING | 260712.1N 0913525.5E | 170 FT | NIL | INRES SITE, SHED TOP |
| In circling area and at AD | POLE | 260712.0N 0913525.8E | 168 FT | NIL | Light Pole |
| In circling area and at AD | POLE | 260707.0N 0913523.5E | 195 FT | LGTD | MET ANTENNA, LIGHTNING ARRESTER TOP - 59.41 M, OBSTRUCTION LIGHT TOP - 58.56 M, WIND DIRECTION INDICATOR TOP - 58.46 M, POLE TOP - 58.29 M, ANTENNA TOP - 57.80 M. |
| In circling area and at AD | POLE | 260705.1N 0913522.9E | 164 FT | NIL | SIGNAL POLE |
| In circling area and at AD | SIGN | 260704.9N 0913522.8E | 166 FT | NIL | SIGN BOARD |
| In circling area and at AD | POLE | 260705.2N 0913523.2E | 167 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | OTHER | 260712.2N 0913525.9E | 165 FT | NIL | ELECTRICAL JUNCTION BOX |
| In circling area and at AD | ANTENNA | 260540.8N 0913443.1E | 190 FT | NIL | ASMGCS, LIGHTNING ARRESTER TOP - 57.80 M, OBSTRUCTION LIGHT TOP - 57.51 M, ANTENNA TOP - 57.50 M. |
| In circling area and at AD | FENCE | 260553.5N 0913449.0E | 167 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260549.7N 0913447.2E | 167 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260548.2N 0913446.5E | 165 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260543.4N 0913444.6E | 165 FT | NIL | SIGN BOARD |
| In circling area and at AD | OTHER | 260546.9N 0913446.3E | 167 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | OTHER | 260553.7N 0913450.0E | 166 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | OTHER | 260602.4N 0913454.2E | 167 FT | NIL | Bird Watcher Hut |
| In circling area and at AD | POLE | 260603.5N 0913453.8E | 163 FT | NIL | SIGNAL POLE |
| In circling area and at AD | SIGN | 260603.6N 0913453.6E | 165 FT | NIL | SIGN BOARD |
| In circling area and at AD | SIGN | 260605.1N 0913454.6E | 164 FT | NIL | SIGN BOARD |
| In circling area and at AD | SIGN | 260605.0N 0913454.0E | 164 FT | NIL | SIGN BOARD |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | FENCE | 260603.2N 0913453.5E | 165 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260605.5N 0913454.5E | 165 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260609.0N 0913456.2E | 165 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | OTHER | 260611.4N 0913457.9E | 167 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | OTHER | 260623.5N 0913503.3E | 167 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | ANTENNA | 260627.6N 0913504.8E | 189 FT | LGTD | ASMGCS, LIGHTNING ARRESTER TOP - 57.71 M, OBSTRUCTION LIGHT TOP - 57.46 M, ANTENNA TOP - 55.99 M. |
| In circling area and at AD | FENCE | 260623.1N 0913502.7E | 165 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260618.8N 0913500.7E | 166 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260617.3N 0913500.0E | 166 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | OTHER | 260636.1N 0913509.4E | 166 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | FENCE | 260639.0N 0913510.0E | 165 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260637.1N 0913509.2E | 164 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260633.2N 0913507.4E | 167 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | OTHER | 260656.8N 0913519.2E | 166 FT | NIL | BIRD WATCHER HUT |
| In circling area and at AD | FENCE | 260654.1N 0913517.1E | 165 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260650.7N 0913515.6E | 165 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260648.4N 0913514.5E | 164 FT | NIL | AIRPORT BOUNDARY FENCE |
| In circling area and at AD | FENCE | 260646.3N 0913513.5E | 165 FT | NIL | AIRPORT BOUNDARY FENCE |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | OTHER | 260547.8N 0913455.5E | 177 FT | NIL | MOBILE ROAD TRAFFIC(ROAD ELEV - 49.10 M + 5M MOBILE TRAFFIC) |
| In circling area and at AD | OTHER | 260549.5N 0913447.3E | 176 FT | NIL | MOBILE ROAD TRAFFIC(ROAD ELEV - 48.63 M + 5M MOBILE TRAFFIC) |
| In circling area and at AD | OTHER | 260627.5N 0913504.9E | 176 FT | LGTD | MOBILE ROAD TRAFFIC(ROAD ELEV - 48.54 M + 5M MOBILE TRAFFIC) |
| In circling area and at AD | OTHER | 260712.5N 0913526.8E | 177 FT | LGTD | MOBILE ROAD TRAFFIC(ROAD ELEV - 48.98 M + 5M MOBILE TRAFFIC) |
| In circling area and at AD | OTHER | 260710.2N 0913531.8E | 177 FT | LGTD | MOBILE ROAD TRAFFIC(ROAD ELEV - 49.03 M + 5M MOBILE TRAFFIC) |
| In circling area and at AD | ANTENNA | 260913.3N 0914605.7E | 1053 FT | LGTD | TV TOWER |
| In circling area and at AD | POLE | 261038.2N 0914603.3E | 880 FT | LGTD | FLAG POLE ON HILL(GANDHI MANDEEP) |
| In circling area and at AD | POLE | 260701.9N 0913519.0E | 182 FT | NIL | POLE TOP |
| In circling area and at AD | OTHER | 260546.4N 0913444.4E | 183 FT | NIL | WINDSOCK |
| In circling area and at AD | OTHER | 260345.7N 0913555.5E | 318 FT | NIL | H.T.TOWER |
| In circling area and at AD | TRANSMISSION LINE | 260946.5N 0914231.7E | 708 FT | NIL | H.T TOWER ON HILL |
| In circling area and at AD | TRANSMISSION LINE | 261010.9N 0914149.1E | 852 FT | NIL | H.T TOWER ON HILL |
| In circling area and at AD | OTHER | 260501.2N 0913846.6E | 732 FT | NIL | H.T.TOWER ON HILL |
| In circling area and at AD | OTHER | 260548.2N 0913455.1E | 173 FT | NIL | GP HUT NEW |
| In circling area and at AD | OTHER | 260209.9N 0913651.1E | 1219 FT | NIL | TREE ON HILL |
| In circling area and at AD | OTHER | 260230.1N 0913723.7E | 1316 FT | NIL | HILL |
| In circling area and at AD | OTHER | 260118.2N 0913712.2E | 1311 FT | NIL | TREE ON HILL |
| In circling area and at AD | OTHER | 260426.0N 0913801.0E | 784 FT | NIL | TREE ON HILL |
| In circling area and at AD | OTHER | 260138.3N 0913905.9E | 1628 FT | NIL | HILL |
| In circling area and at AD | OTHER | 260014.6N 0913904.9E | 1887 FT | NIL | TREE ON HILL |

| In Approach/Take-off/Circling Area and at AD | | | | | |
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| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | OTHER | 260033.8N 0914015.6E | 2028 FT | NIL | HILL |
| In circling area and at AD | OTHER | 260350.9N 0914103.5E | 926 FT | NIL | HILL |
| In circling area and at AD | OTHER | 260205.8N 0914201.5E | 1674 FT | NIL | HILL |
| In circling area and at AD | POLE | 260533.2N 0913449.9E | 178 FT | NIL | ELECTRIC POLE |
| In circling area and at AD | BUILDING | 260534.0N 0913450.1E | 177 FT | NIL | SHED TOP |
| In circling area and at AD | POLE | 260647.3N 0913521.4E | 167 FT | NIL | LIGHT POLE |
| In circling area and at AD | FENCE | 260705.1N 0913522.5E | 169 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | OTHER | 260706.5N 0913523.7E | 176 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV. 48.6 M. +5M) |
| In circling area and at AD | OTHER | 260706.1N 0913529.9E | 177 FT | NIL | MOBILE ROAD TRAFFIC (ROAD ELEV. 48.9 M. +5M) |
| In circling area and at AD | OTHER | 260706.6N 0913519.0E | 201 FT | NIL | WATCH TOWER |
| In circling area and at AD | POLE | 260619.9N 0913459.2E | 173 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260620.2N 0913459.3E | 174 FT | NIL | LIGHT POLE |
| In circling area and at AD | OTHER | 260646.7N 0913522.9E | 179 FT | NIL | WATCH TOWER |
| In circling area and at AD | OTHER | 260639.5N 0913519.7E | 168 FT | NIL | SECURITY HUT |
| In circling area and at AD | FENCE | 260646.7N 0913523.0E | 168 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | FENCE | 260642.9N 0913521.3E | 169 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | POLE | 260633.0N 0913516.6E | 164 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260632.4N 0913516.3E | 164 FT | NIL | LIGHT POLE |
| In circling area and at AD | POLE | 260631.8N 0913516.1E | 165 FT | NIL | Light Pole |
| In circling area and at AD | FENCE | 260634.3N 0913517.2E | 166 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | FENCE | 260637.7N 0913518.8E | 168 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | FENCE | 260640.8N 0913520.2E | 168 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | POLE | 260625.6N 0913518.8E | 231 FT | NIL | FLOOD LIGHT |
| In circling area and at AD | POLE | 260622.0N 0913517.1E | 233 FT | NIL | FLOOD LIGHT |
| In circling area and at AD | BUILDING | 260615.9N 0913514.2E | 266 FT | LGTD | ATC TOWER, LIGHTNING ARRESTER TOP - 81.00 M, ANTENNA TOP - 79.53 M, ANTENNA TOP - 79.51 M,OBSTRUCTION LIGHT TOP - 79.24 M,ANTENNA TOP - 78.53 M,ANTENNA TOP - 78.39 M, GRILL TOP - 77.06 M,ATC TOWER TOP - 75.85 M,HOARDING TOP - 65.39 M,BUILDING TOP - 63.50 |
| In circling area and at AD | BUILDING | 260615.9N 0913514.2E | 254 FT | LGTD | ATC TOWER, AERDROME BEACON |
| In circling area and at AD | OTHER | 260611.2N 0913508.0E | 184 FT | LGTD | WINDSOCK, OBSTRUCTION LIGHT TOP - 56.02 M, WINDSOCK TOP - 55.71 M. |
| In circling area and at AD | OTHER | 260607.0N 0913504.7E | 171 FT | NIL | SECURITY HUT, LIGHT POLE TOP - 52.18 M, SECURITY HUT TOP - 51.49 M. |
| In circling area and at AD | POLE | 260607.0N 0913504.9E | 173 FT | NIL | LIGHT POLE |
| In circling area and at AD | FENCE | 260607.0N 0913504.6E | 168 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | FENCE | 260605.8N 0913504.0E | 170 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | FENCE | 260602.9N 0913502.7E | 170 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | OTHER | 260600.4N 0913501.5E | 168 FT | NIL | ELECTRICAL JUNCTION BOX |
| In circling area and at AD | FENCE | 260549.6N 0913456.4E | 171 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | FENCE | 260547.6N 0913455.5E | 171 FT | NIL | AIRPORT BOUNDARY WALL WITH FENCING ON TOP |
| In circling area and at AD | FENCE | 260544.5N 0913454.1E | 167 FT | NIL | AIRPORT BOUNDARY WALL |
| In circling area and at AD | ANTENNA | 260539.7N 0913451.8E | 191 FT | LGTD | ASMGCS (RU-09), LIGHTNING ARRESTER TOP - 58.25 M, OBSTRUCTION LIGHT TOP - 57.97 M, ANTENNA TOP - 57.95 M. |
| In circling area and at AD | OTHER | 260539.8N 0913451.9E | 172 FT | NIL | SECURITY HUT, LIGHT TOP - 52.47 M, HUT TOP - 51.54 M. |
| In circling area and at AD | GATE | 260539.5N 0913451.7E | 168 FT | NIL | CRASH GATE |
| In circling area and at AD | NAVAID | 260714.2N 0913524.1E | 183 FT | NIL | GAGAN ANTENNA, LIGHTNING ARRESTER TOP - 55.70 M, ANTENNA TOP - 55.34 M, GRILL TOP - 54.24 M. |
| In circling area and at AD | NAVAID | 260714.0N 0913523.5E | 183 FT | NIL | GAGAN ANTENNA, LIGHTNING ARRESTER TOP - 55.65 M, ANTENNA TOP - 55.31 M, GRILL TOP - 54.18 M. |
| In circling area and at AD | NAVAID | 260713.8N 0913523.9E | 183 FT | NIL | GAGAN ANTENNA, LIGHTNING ARRESTER TOP - 55.65 M, ANTENNA TOP - 55.33 M, GRILL TOP - 54.20 M. |
| In circling area and at AD | OTHER | 260620.0N 0913459.3E | 186 FT | LGTD | WINDSOCK |
| In circling area and at AD | TREE | 260637.4N 0913520.8E | 192 FT | NIL | TREE |
| In circling area and at AD | POLE | 260615.4N 0913513.7E | 231 FT | LGTD | FLOOD LIGHT MAST |
| In circling area and at AD | POLE | 260617.2N 0913514.9E | 233 FT | LGTD | FLOOD LIGHT MAST |
| In circling area and at AD | ANTENNA | 260548.9N 0913454.5E | 175 FT | LGTD | GP DME ANTENNA TOP (L) (BASE ELV.=48.7 M) |
| In circling area and at AD | ANTENNA | 260547.0N 0913455.6E | 209 FT | LGTD | GP MONITOR ANTENNA TOP (L) (BASE ELV.=48.6M) |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation (EGM08) | Marking/LGT | Remarks |
| In circling area and at AD | ANTENNA | 260549.0N 0913455.5E | 176 FT | LGTD | GP MAIN ANTENNA TOP (L) (BASE ELV.=48.7 M) |

## VEGT AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

| | | |
|---|---|---|
| 1 | Name of the associated meteorological office | Aerodrome Meteorological Office, Guwahati |
| 2 | Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours | H24 |
| 3 | Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts | Aerodrome Meteorological Office, Guwahati<br>9HR and 24HR |
| 4 | Availability of the trend forecast for the aerodrome and interval of issuance | Trend<br>30 MIN |
| 5 | Information on how briefing and/or consultation is provided | Provided |
| 6 | Types of flight documentation supplied and language(s) used in flight documentation | Tabular Form English |
| 7 | Charts and other information displayed or available for briefing or consultation | S,U85,U70,U50,U30,U20 |
| 8 | Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images; | Telex, Telefax, Satellite Display Work Station Wx Radar. |
| 9 | The air traffic services unit(s) provided with meteorological information | VEGT Guwahati ATC and ACS |
| 10 | Additional information, e.g. concerning any limitation of service. | NA |

## VEGT AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

| Designations | TRUE Bearings | Dimensions of RWY (M) | Strength of pavement (PCR) and associated data) and surface of runway and associated stopways | Geographical coordinates for threshold and runway end |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 02 | 22.72 DEG | 3103 x 45 M | 776/F/C/W/T<br>Asphalt | THR:<br>260541.22N<br>0913447.07E<br>RWY END:<br>260714.23N<br>0913530.21E |
| 20 | 202.72 DEG | 3103 x 45 M | 776/F/C/W/T<br>Asphalt | THR:<br>260714.23N<br>0913530.21E<br>RWY END:<br>260541.22N<br>0913447.07E |

| THR elevation and highest elevation of TDZ of precision APP RWY | Slope of runway and associated stopway | Dimensions of stopway (M) | Dimensions of clearway (M) | Dimensions of strips (M) |
|---|---|---|---|---|
| 6 | 7 | 8 | 9 | 10 |
| THR: 163.0FT<br>TDZ: 163.0FT | | NIL | NIL | 3223 x 150 M |
| THR: 163.0FT<br>TDZ: 163.0FT | | NIL | NIL | 3223 x 150 M |

| Dimensions of runway end safety areas | Location and description of arresting system (if any) | Existence of an obstacle-free zone | Remarks. |
|---|---|---|---|
| 11 | 12 | 13 | 14 |
| 240M x 90M | | NIL | Vertical Datum: EGM 08 |
| 90M x 90M | | NIL | Vertical Datum: EGM 08 |

## VEGT AD 2.13 DECLARED DISTANCES

| RWY Designator | Take-off run available TORA (M) | Take-off distance available TODA (M) | Accelerate distance available ASDA (M) | Landing distance available LDA (M) | Remarks (including runway entry or start point where alternative reduced declared distances have been declared) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 02 | 3103 | 3103 | 3103 | 3103 | TORA from TWY L2: 2371 M<br>TORA from TWY F: 1760 M<br>TORA from TWY E: 1566 M<br>TORA from TWY L3: 1072 M |
| 20 | 3103 | 3103 | 3103 | 3103 | TORA from TWY L4: 2734 M<br>TORA from TWY L3: 2031 M<br>TORA from TWY E: 1537 M<br>TORA from TWY F: 1343 M |

## VEGT AD 2.14 APPROACH AND RUNWAY LIGHTING

| Runway Designator | Type, length and intensity of approach lighting system | Runway threshold lights, colour and wing bars | Type of visual slope indicator system | Length of runway touchdown zone lights |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 02 | CAT I<br>570 M<br>LIH | Green | PAPI<br>LEFT/3.25 DEG<br>MEHT<br>(54.00FT) | |
| 20 | | Green | PAPI<br>LEFT/3.00 DEG<br>MEHT<br>(52.16FT) | |

| Length, spacing, colour and intensity of runway centre line lights | Length, spacing, colour and intensity of runway edge lights | Colour of runway end lights and wing bars | Length and colour of stopway lights | Remarks |
|---|---|---|---|---|
| 6 | 7 | 8 | 9 | 10 |
| | 3103 M<br>60 M<br>White<br>LIH | Red | | 1.RWY Edge light variable white and last 600m Yellow in approach direction.<br>2.PAPI RWY 02 commissioned at a distance of 323m from threshold and 15m fm RWY edge, left, MEHT 54ft, glide angle 3.25 deg. |
| | 3103 M<br>60 M<br>White<br>LIH | Red | | RWY Edge light variable white and last 600m Yellow in approach direction. |

## VEGT AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

| | | | |
|---|---|---|---|
| 1 | Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any) | ABN | At Tower Building, FLG W & G Ev 2 SEC,<br>As ATS HR |
| | | IBN | Nil |
| 2 | Location and lighting (if any) of anemometer/ landing direction indicator; | LDI | 100M NW OF ATC TOWER. |
| | | Anemometer | On tower building lighted |
| 3 | Taxiway edge and taxiway centre line lights; | Edge | All TWY. |
| | | Centre Line | |
| 4 | Secondary power supply including switch-over time; | Secondary Power supply to all lighting at AD.<br>Switch-over time: 15 SEC. | |
| 5 | Remarks | NIL | |

## VEGT AD 2.16 HELICOPTER LANDING AREA

| | | |
|---|---|---|
| 1 | Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area | Not Established |
| 2 | TLOF and/or FATO area elevation: | Not Established |
| 3 | TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking; | Not Established |
| 4 | True bearings of FATO; | Not Established |
| 5 | Declared distances available | Not Established |
| 6 | Approach and FATO lighting; | Not Established |
| 7 | Remarks | Not Established |

## VEGT AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

| | | |
|---|---|---|
| 1 | Airspace designation, geographical coordinates and lateral limits | CTR: Area bounded by lines joining points 262702N 0921448E; 260737N 0922332E; 260715N 0922155E; 254720N 0921620E; 253502N 0922948E; 253502N 0905949E; 262702N 0905949E to point of origin. |
| 2 | Vertical limits | FL 85 |
| 3 | Airspace classification | D |
| 4 | Call sign and language(s) of the air traffic services unit providing service; | Guwahati Tower / Guwahati Approach, English |
| 5 | Transition altitude | 9000 FT |
| 6 | Hours of applicability | HO |
| 7 | Remarks | NIL |

## VEGT AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

| Service Designation | Call sign | Channel(s) | SATVOICE Number(s), if available |
|---|---|---|---|
| **1** | **2** | **3** | **4** |
| TAR | Guwahati Radar | 123.900 MHZ | |
| TAR | Guwahati Radar | 127.950 MHZ | |
| SAR | - | 123.100 MHZ | |
| APP | Guwahati Approach | 123.900 MHZ | |
| TWR | Guwahati Tower | 118.750 MHZ | |
| TWR | Guwahati Tower/Guwahati Ground | 122.050 MHZ | |
| ATIS | Guwahati Information | 126.600 MHZ | |
| ACC | Guwahati Control | 120.500 MHZ | |
| ACC | Guwahati Control | 128.775 MHZ | |
| ALRS | Guwahati Emergency | 121.500 MHZ | |
| RADAR | Guwahati Radar | 120.500 MHZ | |
| SMC | Guwahati Ground | 121.900 MHZ | |

| Logon address, as appropriate | Hours of operation | Remarks |
|---|---|---|
| **5** | **6** | **7** |

| Logon address, as appropriate | Hours of operation | Remarks |
|---|---|---|
| | As ATS | Main |
| | As ATS | SDBY |
| | As ATS | NIL |
| | AS ATS | MAIN |
| | As ATS | NIL |
| | HO | Backup Frequency for TWR/GND |
| | As ATS | NIL |
| | As ATS | NIL |
| | | SDBY FREQUENCY |
| | AS ATS | NIL |
| | As ATS | NIL |
| | As ATS | NIL |

## VEGT AD 2.19 RADIO NAVIGATION AND LANDING AIDS

| Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid | Identification | Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate | Hours of operation, as appropriate; |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| LOC 02<br>CAT I | IGHT | 110.300 MHz | HO |
| GP 02 | | 335.000 MHz | As ATS |
| DME ILS 02 | IGHT | CH40X | HO |
| DVOR/DME | GGT | 117.600 MHz<br>CH123X | H24 |

| Geographical coordinates of the position of the transmitting antenna | Elevation of transmitting antenna of DME/ elevation of GBAS reference point | Service volume radius from the GBAS reference point | Remarks |
|---|---|---|---|
| 5 | 6 | 7 | 8 |
| 260739.9N<br>0913542.1E | | | |
| 260549.0N<br>0913455.5E | | | Glide angle 3.25 DEG |
| 260548.9N<br>0913455.6E | 187 FT | | Gravity Model: EGM08 |
| 260802.9N<br>0913552.5E | 193 FT | | Elevation in EGM 08 |

## VEGT AD 2.20 LOCAL AERODROME REGULATIONS

### I. AIRPORT COLLABORATIVE DECISION MAKING (A-CDM)

1.1 The Lokpriya Gopinath Bordoloi Airport, Guwahati - Collaborative Decision Making (A-CDM, Guwahati) undertaken at Guwahati International Airport is a joint programme among all airport partners –

- Air Navigation Service Provider (ATC)
- Aircraft Operators (AO)
- Airport Operator/AOCC
- Ground Handling Agencies (GHA)
- Central Air Traffic Flow Management Unit (CATFMU), New Delhi
- Air Force Movement Liaison Unit (AF-MLU), Kolkata
- MLC Guwahati

All the partners are required to work in close collaboration to ensure the successful operation of A-CDM, Guwahati.

1.2 The efficiency of the Air Transport System is highly dependent on traffic predictability. A-CDM, Guwahati effectively enhances predictability (this reduces buffer times for resource planning and flight times), overall efficiency and punctuality by linking and sharing of accurate and timely information amongst Aircraft Operators, Airport Operator, ATC, etc.

1.3 A-CDM application is developed in-house by Airports Authority of India.

1.4 A-CDM system processes and utilizes the following information from the various sources mentioned below:

a) AFTN:
   i) Flight plan data
   ii) EOBT
   iii) ATD
   iv) ETAs
   v) NOTAM
   vi) METARs
   vii) ADC Numbers

b) Airport Operations Data Base (AODB) of Guwahati
   i) Parking stand of aircraft
   ii) Registration marking
   iii) Boarding status of departing aircraft
   iv) Actual “Off Block” Times (AOBT)
   v) Actual “In-Block” Times (AIBT)

c) A-CDM system database
   i) Arrival and Departure Taxi timings from each parking stand
   ii) Runway-in-Use and Closures
   iii) Hourly handling capacity of runways
   iv) Priority of flights e.g. VVIP and Emergencies etc.
   v) Flight schedules

1.5 A-CDM is a Common Information Sharing Platform introduced to process timely and accurate information for quick and precise decision making. A-CDM takes into consideration the ETAs of arriving aircraft, Target Off Block Times (TOBT) of departures, Runway-in-use and the handling capacity of the runway and determines the arrival and departure sequence. Based on the sequence and the parking position, it calculates the Target Take-Off Times (TTOT) & Target Start up Times (TSAT) of departures and Target In-Block Times (TIBT) for arrivals.

1.6 Implementation of ACDM will result in considerable benefits to stakeholders. The various benefits of A-CDM includes are as follows:

• Improvement in efficiency of operations

• Reduction of delays to departures at holding point.

• Orderly flow of traffic.

• Reduction in wastage of Aviation fuel.

• Reduction in Carbon emmission, reduction in Environmental Pollution.

• Optimization of airport and airspace capacity

• Reduction of R/T congestion

• Enhanced efficiency and reduced workload due to reduced R/T congestion.

• Improved Situational Awareness and increased confidence amongst ACDM Partners.

• Adherence to ADC (Air Defence Clearance) by stake holders.

• Better Passenger experience.

• Optimized use of ground handling resources

• Optimized use of parking stands, gates and terminals.

• Reduced apron and taxiway congestion

• Access to current MET information.

## 2. TWO KEY PARAMETERS OF A-CDM, GUWAHATI: TOBT and TSAT.

### 2.1 TARGET OFF-BLOCK TIME (TOBT)

**2.1.1 Definition:** The time that an Aircraft Operator estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up/push back immediately upon reception of clearance from the Surface Movement

**Control Unit (SMC)**. It is the real time target of ensuring readiness of an aircraft and therefore, more accurate than the static departure time mentioned in the flight plan and hence, an ideal milestone to be used by all airport partners.

**2.1.2 Purpose:** Air Traffic Management (ATM) based on only FPL-data is insufficient to guarantee smooth and efficient operations on the ground and in the air. The main reason for this is the incorrect updating/no updating at all, of the Estimated Off Block Time (EOBT) of the aircraft. The result is that airport partners will have an incorrect idea about the actual status of that specific aircraft. The purpose of Target Off-Block Time (TOBT) is to provide a timely, accurate and reliable estimate of an aircraft's off block time to the A-CDM, Guwahati partners. Accurate TOBTs enhance operations on the ground as they provide all airport partners with a clear picture of the intentions of aircraft on the ground.

**2.1.3 Communication of TOBT:** Aircraft Operators shall revise their EOBTs via AFTN using CHG/ DLA message or inform ARO for revised EOBT, who in turn will originate CHG / DLA message for transmission through AFTN.

**2.1.3.1 In case No delay is expected:** If there is no revision of EOBT, EOBT will be treated as the TOBT. For such cases there is no need of any communication from the Aircraft Operator to the ARO.

**2.1.3.2 In case delay is expected:**

**a) When Air Defence Clearance (ADC) is valid**: If delay is within the validity period of ADC, EOBT should be revised through CHG/DLA message. Optionally TOBT can be changed in ACDM interface. But if due to some reason coordination is not possible by both the above methods then EOBT can be revised by informing ARO.

**b) When Air Defence Clearance ADC is NOT valid:**

i) If delay is outside the validity period of ADC, EOBT shall be revised through CHG/DLA message or after having coordinated with MLU and obtaining the revised ADC number, by informing ARO about revised EOBT and ADC. (Option of change of TOBT will not be available in this case).

ii) ADC Validity Period: (-) 15 minutes to (+) 45 minutes of EOBT in RPL/FPL or subsequent revision of EOBT by DLA/ CHG message.

**Note: Once Target Start-up Approval Time (TSAT) is allocated by ATC at TOBT - 30 minutes, the TOBT can be updated thrice.**

**2.1.4 Accuracy:** TOBTs must be updated to an accuracy of +/- 5 minutes.

**2.2 TARGET START UP APPROVAL TIME (TSAT)**

**2.2.1 Definition:** The time is calculated by A-CDM application taking into account TOBT, CTOT* and/or the traffic situation that an aircraft can expect to start-up/push back for departure.

Note: In case there is no congestion, ATC may allow the aircraft to start, where Actual Start-up Approval Time (ASAT) will be before the TSAT.

**2.2.2 Purpose:** TSATs provide an optimized start-up sequence, leading to Pre-Departure sequence. TSATs reduce queuing times at the runway holding points, while ensuring optimised utilisation of the runway. ATC will continue to optimize the departure order for creating the right mix of traffic.

**2.2.3 Calculation of TSAT:** TSAT is calculated automatically by A-CDM Application. The TSAT is calculated by taking into account TOBT, priority of aircraft, variable taxi times (VTT), and the runway capacity. Any revision of EOBT will trigger the A-CDM application to recalculate a new TSAT for that flight as well as to re-allocate earlier TSAT to next flight which meets the requisite conditions. TSAT is displayed in various ACDM interfaces in the following colour coded information.

Blue colour: TSAT - 15 minutes up to TSAT - 5 minutes;

Green colour: TSAT - 5 minutes up to TSAT + 5 minutes;

Red colour: After TSAT + 5 minutes.

**2.2.4 Generation of TSAT:** TSAT is generated by ACDM interface at TOBT - 30 minutes.

**2.2.5 Communication of TSAT:** TSAT will be displayed in various A-CDM interfaces. The AO/GHA shall advise flight crew of TSAT, displayed in the A-CDM interface.

**2.2.6 Priorities for generation of TSAT**:

The following aircraft will be accorded priority by A-CDM application at the time of generating TSATs:

a) VVIP flights

b) Aircraft in emergency

c) Ambulance aircraft

d) Aircraft in relief missions

e) Aircraft returning to stand after push back/taxiing out for any reason.

**2.2.7** Airlines/GHAs are required to monitor the A-CDM interface regularly to get information on the revised TSAT if any, in respect of their aircraft. It may please be noted that the earlier allocated TSAT may get changed due to the following reasons:

i. Change of runway

ii. Change in taxi times

iii. Revised TOBT of the subject aircraft

iv. Cancellation/ revision in TOBT of other aircraft in the departure sequence.

v. Priority handling

vi. Unusual Occurrences, etc.

vii. Change of aircraft type*

viii. Application of ATFM regulation or ATC restriction resulting in new CTOT or start up delay*

ix. Inclement Weather*

x. Bird activity*

xi. Change of route (SID)*

**2.2.8 Accuracy:** The TSAT will remain valid for +/- 5 minutes.

**3 CO-ORDINATION WITH THE CENTRAL AIR TRAFFIC FLOW MANAGEMENT UNIT (CATFMU)***

**3.1** A permanent and fully automatic data exchange with the ATFMU will be established. The data transfer will enable highly accurate early predictions of landing and departure times. *

**3.2 Furthermore, it allows for more accurate and efficient CTOT calculations due to the use of more accurate local target take-off times. ***

**4. TOBT AND TSAT BASED A-CDM, GUWAHATI START-UP/PUSH-BACK PROCEDURES**

**4.1 Pre-Departure Sequence:** Pre-Departure sequence is the order, in which the ATC plans the aircraft to depart from their Gate/Stands. It should not be confused with the Pre-Take-off sequence where ATC decides on the order in which the aircraft at holding points of the runway will depart.

**4.2 Start-Up and Push-Back Procedure:**

i. The pilot will contact Surface Movement Control Unit (GROUND) to request en-route clearance and SID between TSAT - 15 minutes to TSAT - 5 minutes. Blue Zone.

ii. The aircraft must be ready to Start-up/Push-back at TOBT and request SMC (GROUND) for Start-up/Push-back at TSAT – 5 minutes and TSAT + 5 minutes. Green Zone.

(Note: The system may prepone TSAT up to TOBT)

iii. In case for Schedule Flight, “Boarding” has not started at or before TSAT – 5 Minutes, allotted TSAT will be cancelled and associated slot will be allocated to next eligible aircraft.

a. A new TOBT is required from Airlines Operator.

b. On receipt of new TOBT, the flight will be re-sequenced according to new TOBT and a new TSAT with a subsequent delay will be issued.

iv. **If at TSAT +5 minutes: Red Zone, ATC has not received Start-up / Pushback request, the aircraft will lose its position in sequence.**

a. A new TOBT is required from Airlines Operator.

b. On receipt of new TOBT, the flight will be re-sequenced according to new TOBT and a new TSAT with a subsequent delay will be issued.

v. ATC should normally be able to issue start-up/push-back at TSAT. Pilots will be informed by ATC of any revised TSAT if there is a delay to TSAT in excess of 5 minutes.

vi. Taxi clearance must be requested within 5 minutes of Start-up/Pushback approval time.

(Note: If this has not occurred, SMC (GROUND) must be notified of the extent of delay. In such cases, aircraft may lose its departure slot and a new TOBT may be required in A-CDM, Guwahati interface. AO shall be responsible to obtain new ADC number from IAF-MLU).

vii. The Pre-Departure (Off- block) Sequence will be determined in accordance with Target Start up Approval Time (TSAT) and NOT in accordance with the Start-up Request. Pre-Departure Sequence will not have any bearing on Actual Departure sequence.

(Note: Actual Departure sequence may differ from pre-departure sequence in order to optimize the Runway and Airspace utilization.)

* [Future provision kept for incorporating with India – Central Air Traffic Flow Management]

5 TERMS AND ABBREVIATIONS

All Acronyms are time parameters which have a standard length of four characters and have been arranged as per aircraft movement sequence.

[While developing the A-CDM, Guwahati procedures, the guidance material published by Euro control, ICAO A-CDM and other Organizations has been used].

| S.NO | ACRONYMS | DEFINITION | EXPLANATION |
|---|---|---|---|
| 1 | ELDT | Estimated Landing Time | The estimated time that an aircraft will touchdown on the runway. (Equivalent to ATC ETA = Estimated Time of Arrival = landing). |
| 2 | ALDT | Actual Landing Time | The time that an aircraft lands on a runway. (Equivalent to ATC ATA = Actual Time of Arrival = Landing, ACARS = ON). |
| 3 | EXIT | Estimated Taxi- In Time | The estimated taxi time between landing and in-Block. |
| 4 | AXIT | Actual Taxi-In Time | AXIT = AIBT – ALDT |
| 5 | SIBT | Scheduled In- Block Time | The time that an aircraft is scheduled to arrive at its first parking position |
| 6 | EIBT | Estimated In- Block Time | The estimated time that an aircraft will arrive in-blocks. (Equivalent to Aircraft Operator's ETA = Estimated Time of Arrival). |
| 7 | AIBT | Actual In-Block Time | The time that an aircraft arrives in-blocks. (Equivalent to Aircraft Operator's ATA = Actual Time of Arrival, ACARS = IN). |
| 8 | STTT | Scheduled Turnaround Time | STTT = SOBT – SIBT |
| 9 | ETTT | Estimated Turnaround Time | The time estimated by the AO/GH on the day of operation to turnaround a flight taking into account the operational constraints |
| 10 | MTTT | Minimum Turn-around Time | The minimum turnaround time agreed with an AO/GH for a specified flight or aircraft type. |
| 11 | ATTT | Actual Turn-around Time | ATTT = AOBT – AIBT |
| 12 | ASRT | Actual Start Up Request Time | Time the pilot requests start up clearance |

| | | | |
|---|---|---|---|
| 13 | TSAT | Target Start Up Approval Time | The time provided by ATC taking into account TOBT, CTOT* and/or the traffic situation that an Aircraft can expect start-up / push back approval. Note: The actual start up approval (ASAT) can be given in advance of TSAT. |
| 14 | ASAT | Actual Start Up Approval Time | Time that an aircraft receives its start-up approval. |
| 15 | SOBT | Scheduled Off- Block Time | The time that an aircraft is scheduled to depart from its parking position |
| 16 | EOBT | Estimated Off- Block Time | The estimated time at which the aircraft will start movement associated with departure (ICAO). |
| 17 | TOBT | Target Off-Block Time | The time that an Aircraft Operator estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up / push back immediately upon reception of clearance from the SMC (GROUND). |
| 18 | AOBT | Actual Off-Block Time | Time the aircraft pushes back / vacates the parking position. (Equivalent to Aircraft Operator's ATD = Actual Time of Departure & ACARS = OUT) |
| 19 | ARDT | Actual Ready Time (for Movement) | When the aircraft is ready for start-up/push back or taxi immediately after clearance delivery, meeting the requirements set by the TOBT definition |
| 20 | EXOT | Estimated Taxi- Out Time | The estimated taxi time between off-block and take off. This estimate includes any delay buffer time at the holding point prior to take off. |
| 21 | AXOT | Actual Taxi-Out Time | AXOT = ATOT – AOBT |
| 22 | ETOT | Estimated Take Off Time | The estimated take off time taking into account the EOBT plus EXOT. |
| 23 | CTOT* | Calculated Take Off Time* | A time calculated and issued by the appropriate Central Management unit, as a result of tactical slot allocation, at which a flight is expected to become airborne. (ICAO Doc 7030/4 – EUR, Table 7)* |
| 24 | TTOT | Target Take Off Time | The Target Take Off Time taking into account the TOBT /TSAT plus the EXOT. Each TTOT on one runway is separated from other TTOT or TLDT to represent vortex and/or SID separation between Aircraft. |
| 25 | ATOT | Actual Take Off Time | The time that an aircraft takes off from the runway. (Equivalent to ATC ATD = Actual Time of Departure, ACARS = OFF). |

II. Taxing of aircraft on TWY A not permitted behind an aircraft holding at RWY Holding Position on TWY G for RWY 20.
III Taxiing of Code D or E aircraft on TWY A toward Apron I, beyond the junction with TWY G, NOT PERMITTED.
IV. Arriving Code D/E aircraft shall vacate RWY via TWY G, then Turn left to join TWY A for Stand 21, or Proceed straight on TWY G toward Apron K or Isolated Aircraft Parking Position (IAPP).
V. Departing aircraft from 21: Taxi via TWY A, right turn to join TWY G for RWY 20 entry.
VI. Departing Aircraft from Apron K or IAPP: Taxi on TWY G to RWY holding position. Left turn onto TWY A not permitted.
VII. Deployment of wing walkers/wing guards by airline is mandatory prior to undertaking pushback from any stand at LGBI Airport or during aircraft arrival at stand.

**VIII. PUSHBACK AND START UP PROCEDURES FOR MULTIPLE AIRCRAFTRECEIVING STAND (MARS) 21L,21, 21R, AND AIRCRAFT STANDS 22L, 22R, 23L, 27 AND 28, AND ASSOCIATED DEPENDENCIES FROM OTHER AIRCRAFT STANDS:**

| Stand | Pushback Procedure | Caution | Reference |
|---|---|---|---|
| 21L, 21R | • Pushback facing North North-East on Taxilane D1 up to Tug Release Point Z1<br>• Taxi out via Taxilane D1 -- TWY C8 | Pushback from stands 21, 21L, 21R, 22L, 22R and 23L are interdependent. | Refer Aircraft Pushback Procedure Chart |
| 21 | • Pushback facing North North-East on Taxilane D up to Tug Release Point Z2<br>• Taxi out via Taxilane D -- TWY C8 | Pushback from stands 21, 21L, 21R, 22L, 22R and 23L are interdependent. | Refer Aircraft Pushback Procedure Chart |
| 22L, 22R, 23L | Pushback facing North North-East on Taxilane D1 and pull forward up to Tug Release Point Z1<br>•Taxi out via Taxilane D1 -- TWY C8 | Pushback from stands 21, 21L, 21R, 22L, 22R and 23L are interdependent. | Refer Aircraft Pushback Procedure Chart |
| 27, 28 | •Pushback facing North North-East on Taxilane D3 and pull forward up to Tug Release Point Z3<br>•Taxi out via Taxilane D3 -- TWY C8 | Pushback from stands 27, 28 are interdependent. | Refer Aircraft Pushback Procedure Chart |

**VEGT AD 2.21 NOISE ABATEMENT PROCEDURES**

NIL

## VEGT AD 2.22 FLIGHT PROCEDURES

1.GENERAL

i.Radar vectoring shall be provided to intercept final approach track prior to 12 NM from touch down at 3500FT. Final approach track for ILS and VOR DME procedure is 023 DEG M and 023 DEG M (R-203) respectively.

ii.Radar service shall be terminated when aircraft is established on final approach track and not earlier than 11 miles from touch down.

iii.Radar controller shall give " 20 miles from touch down " position report to the aircraft.

iv.When radar vectoring is provided, aircraft speed shall be restricted as given below: -
a)220 KTS IAS maximum on passing F100.
b)185 KTS IAS maximum when 20 NM from touch down position information is given by radar controller.

v.In case of ILS procedure final approach descent shall commence on interception of glide path at 3000FT.

vi.ILS with Glide path inoperative procedure shall be applicable for aircraft equipped with ILS DME. Aircraft shall leave 3000FT for 2300FT at 12D (ILS DME) after interception of localizer (110.3 IGHT) inbound track 023 DEG M.

vii.In case of VOR DME procedure RWY 02 aircraft shall leave 3500FT for 2800FT at 16D (VOR DME) when established on final approach track 023 DEG M (R-203)

viii.When being vectored via predetermined tracks aircraft shall arrange the descent profile in such a way, so as to reach 3500FT prior 12NM from touch down on final approach track.

2.RADAR VECTORING PROCEDURES

i.ATS Route R472
At 30 NM from MSSR aircraft may be descended to 5000FT. At 25NM aircraft shall be radar vectored to fly predetermined track 090 DEG M. On passing 230 DEG M/20NM from MSSR aircraft shall be descended to 3500FT and after crossing 214 DEG M from MSSR given left turn to follow track 065 DEG M to intercept the final approach track.

ii.ATS Route W137
At 30 NM, aircraft shall be given radar vector to fly predetermined track 141 DEG M descending to 6000FT. At 233 DEG M/ 19 NM from MSSR aircraft shall be given left turn to fly track 090 DEG M as specified for ATS Route R472 descending to 3500FT and thereafter procedure as given in (i) above for interception of final approach track shall be followed.

iii.ATS Route W53
At 25NM, aircraft shall be descended to 6000FT. At 18NM aircraft shall be vectored to fly predetermined track 270 DEG M descending to 5000FT. Passing 169 DEG M/13NM from MSSR aircraft shall be given descent to 3500FT and thereafter crossing bearing 195 DEG from MSSR given right turn on heading 340 to intercept the final approach track.

iv.ATS Route B593
At 25NM aircraft shall be radar vectored to fly predetermined track 325 DEG M descending to 7000FT/6000FT/5000FT as appropriate as per minimum vectoring altitudes. At 214 DEG M/18NM from MSSR aircraft shall be directed to turn right to follow track 065 DEG M as specified for ATS Route R472 and descended to 3500FT to intercept the final approach track.

v.ATS Route W51
At 25NM, aircraft shall be radar vectored via predetermined track 220 DEG M descending to 6000FT. At 147 DEG M/15NM from MSSR aircraft shall be given right turn to follow predetermined track 270 DEG M descending to 5000FT. Thereafter procedure as given in (iii) for ATS route W53 shall be followed.

vi.Basic Radar Circuit Runway 02
Aircraft shall be radar vectored from 9NM abeam the MSSR on to predetermined track 205 DEG M for left hand down wind and descended to 6000FT. When on track 205 DEG M aircraft shall be directed to turn left at 233 DEG/18NM from MSSR to follow track 090 DEG M as specified for ATS route R472 descending to 3500FT and thereafter procedure as explained in 2(i) for interception of final approach track of ILS or VOR DME approach shall be followed.

3.Radar controller may at his discretion, provide radar vectoring when aircraft is within Guwahati TMA for establishing it on the predetermined tracks, provided interception of final approach track takes place prior to 12NM from touch down at 3500FT.

NOTE: -
a)Care has been taken, while designing the procedures, to provide 5NM lateral clearance between the obstacles and the intended position of the aircraft following the procedures prior to commencement of the final approach. Controllers and pilots shall strictly adhere to the laid down procedures.
b)Aircraft operating VFR between Guwahati and Shillong/South sector must follow via Dispur to avoid Approach funnel of RWY 02. All aircraft operating in VEGT FIR/TMA should keep transponder on. All helicopters from RANGIA to GUWAHATI to route as follows: - RANGIA – 209 DEG 12.5 NM – ALPHA – 180 DEG 8.7 NM – BRAVO – 090 DEG 5 NM – GGT GUWAHATI.

4. ADS-B based ATS Surveillance services are provided to suitably equipped aircraft on opportunity basis within the coverage area of Guwahati, Dibrugarh and Agartala ADS-B sensors in the jurisdiction of Guwahati Area control.

ADS-B based ATS Surveillance services are also provided to suitably equipped aircraft on opportunity basis within the coverage area of Guwahati ADS-B sensors in the jurisdiction of Guwahati Approach control.

ADS-B equipped aircraft flying within Guwahati Area Control and Approach Control to comply with relevant provisions.

**RADIO COMMUNICATION FAILURE PROCEDURES**

**1.INTRODUCTION:**

The Radio Communication Failure procedures established herein are based on the provisions of ICAO PANS ATM DOC 4444. The objective is to standardize the actions to be taken by the pilot of arriving and departing aircraft experiencing radio communication failure at Guwahati airport, when ATS surveillance system is used in the provision of air traffic control in such airspace.
During the periods of non-availability of ATS surveillance system in part or whole of the airspace concerned in Guwahati, the provisions of ICAO PANS ATM DOC 4444 shall be followed.

**2.SCOPE:**

The procedures in this document shall be applicable to the pilots of arriving and departing aircraft experiencing radio communication failure at LGBI Airport, Guwahati when ATS surveillance system is used in the provision of air traffic control in such airspace.

**3.GENERAL:**

3.1 All Transponder equipped aircraft experiencing Radio Communication Failure shall set transponder to Mode A/C code 7600 as soon as practicable. (Note: This Requirement of Setting transponder to Mode A/C code 7600 in no way imposes any restriction on the pilot's decision to set transponder to Mode A/C code 7500 or 7700, whenever required).
3.2 An aircraft equipped with other surveillance system transmitters, including ADS-B might indicate the loss of air-ground communication by all the available means.
3.3 Immediately after detection of RCF, the pilot shall try to communicate to the concerned ATS unit through the frequencies of adjacent units/stations, or 121.5 MHz or by relay through other aircraft.
3.4 Pilot shall make blind transmission to ATC of all necessary reports and actions taken by the aircraft, e.g. descent, turn, proceeding to waypoint etc., irrespective of whether partial/ complete RCF has been established or not.

**4.ASSIGNED RUNWAY AND ITS AVAILABILITY FOR RCF AIRCRAFT:**

4.1 In case of an arriving aircraft experiencing RCF, when Runway for landing has already been advised to the aircraft by ATC, such runway shall be considered as the assigned runway. If an arriving aircraft has not been advised of any runway, Runway 02 shall be considered as assigned runway for such arrival.
4.2 In case of departures from Guwahati returning due to RCF, the departure runway of such aircraft shall be considered as assigned runway for landing.
4.3 Runway lights and Approach lights in 'SWITCHED ON' position shall indicate the availability of such Runway for aircraft experiencing RCF.
4.4 Runway lights and Approach lights in 'SWITCHED OFF' position shall indicate non-availability of such Runway for aircraft experiencing RCF.
4.5 In case strong tail wind conditions are encountered during approach for the assigned Runway, the aircraft experiencing RCF will carry out a missed approach on assigned runway and after following complete missed approach procedure for such approach, aircraft will carry out the either visual approach or laid down procedure for the runway which is in opposite direction to the assigned Runway, e.g. Runway 20 in case of assigned Runway is 02.

**5. PROCEDURE FOR ARRIVALS:**

**5.1Arriving Aircraft Being Radar Vectored**

An aircraft which is radar vectored, on experiencing radio communication failure, shall:

| Phase of flight | Actions to be followed by pilot |
|---|---|
| Radio communication failure takes place prior to interception of final approach track | • Select MODE A/C 7600<br>• Maintain the last assigned level or 4000 FT whichever is higher and proceed to GGT VOR via shortest route.<br>• Descend in GGT hold to 4000ft (as applicable) and carry out the laid down procedure for assigned runway |
| Radio communication failure takes place after interception of final approach track | • Continue the approach and land<br>Or<br>• Go around and carry out the missed approach procedure and join the GGT VOR holding at 4000 FT.<br>• 2 minutes after selecting MODE A/C 7600 carry out the laid down procedure for the assigned runway. |

**5.2 Arriving Aircraft – STAR Not Assigned**

When STAR has not been assigned to an arrival experiencing radio communication failure, it shall:

a) Select MODE A/C 7600.AD

b) Descend to the last assigned level and proceed to GGT VOR.

c) At 30 NM GGT VOR descend to 9000 FT.

d) Descend in GGT hold to 4000ft and carry out the laid down procedure for assigned runway

**5.3 Arriving Aircraft – STAR Assigned**

When STAR has been assigned to an arrival experiencing radio communication failure, it shall:

a) Select MODE A/C 7600 and continue the assigned STAR following all level and speed restrictions applicable to STAR, as far as practicable.

b) At the end of the STAR make a convenient turn to intercept localizer or final approach track of the runway in use.

c) If additional track miles are required to adjust descent, hold over the waypoints as per the following table and descend to minimum holding altitude:

| SL. NO | STAR | WAYPOINT | MNM HOLDING ALT |
|---|---|---|---|
| 01 | ATOGA1A | ORIDA | 6000 FT |
| 02 | OPIMO1A | | |
| 03 | IBAPA1A | | |
| 04 | ATOGA1C | VIKEG | 6000 FT |
| 05 | OPIMO1C | | |
| 06 | VENUM1A | PEDKI | 7000 FT |
| 07 | RUPAK1A | | |
| 08 | VENUM1C | GT483 | 3500 FT |
| 09 | RUPAK1C | | |
| 10 | IBAPA1C | | |

**6. PROCEDURE FOR DEPARTURES:**

**6.1 Departure Intending to Continue to Destination**

A departing aircraft experiencing radio communication failure and intending to continue to its filed plan destination shall:

a) maintain the last assigned speed and level, or minimum flight altitude if higher, for a period of 7 minutes following:

i) the time the last assigned level or minimum flight altitude is reached; or

ii) the time the transponder is set to code 7600; or the ADS-B transmitter is set to indicate the loss of air-ground communications; or

iii) the aircraft's failure to report its position over a compulsory reporting point;

Whichever is later and thereafter adjust level aAD 2.1.22.4nd speed in accordance with the filed flight plan.

b) when being vectored or having been directed by ATC to proceed offset using RNAV without a specified limit, proceed in the most direct manner possible to re-join the current flight plan route no later than the next significant point, taking into consideration the applicable minimum flight altitude.

**6.2 Departure Intending to Land Back at Guwahati**

Aircraft experiencing RCF after departure, deciding to return to Guwahati for landing shall follow the following procedure:

| Phase of flight | Actions to be followed by pilot |
|---|---|
| Immediately after departure (no contact with approach control radar) | • Squawk 7600.<br>• Climb to 7000 FT.<br>• Maintain 7000FT on SID or as per heading/track last issued and acknowledged.<br>• 2 minutes after setting squawk 7600 climb to 9000FT.<br>• Maintain 9000FT for further 2 mins. Thereafter, make shorter arc to come over GGT. Join GGT hold.<br>• Descend to 4000 FT in hold and leave GGT follow the laid down procedure for assigned runway. |
| RCF after establishing contact approach control radar | • Squawk 7600.<br>• If below 9000 FT, climb and maintain 9000FT for 2 mins on SID or as per heading/ track last issued and acknowledged. Thereafter, make shorter arc to come over GGT. Join GGT hold.<br>• If above 9000 FT, continue on current clearance until 2 minutes after setting code 7600. Thereafter, stop climb and make a shorter arc to join GGT hold descending to 9000FT.<br>• Descend to 4000 FT in hold and leave GGT follow the laid down procedure for assigned runway. |
| Aircraft intending to return to Guwahati and requiring fuel jettisoning/burning of fuel | • Aircraft requiring to jettison the fuel will join the Hold at 9000FT and after one hold overhead will proceed on R-260 GGT VOR. Hold on R-260 ,30NM from GGT at 9000 FT, Left Hand Pattern, one minute outbound to jettison fuel/burn fuel.<br>• After jettisoning/burning of the fuel aircraft will proceed via R260 to join hold over GGT at 9000 FT. Descend to 4000 FT in the hold and then leave GGT for the laid down procedure for assigned runway |

**7. Procedures for VFR Traffic:**

7.1 In case of a VFR flight experiencing RCF and does not intend to land at Guwahati it shall continue to fly in VMC, land at the nearest suitable aerodrome/ helipad, and report its arrival by the most expeditious means to the ATC, Guwahati.

7.2 In case of a VFR flight experiencing RCF and intending to land at Guwahati:

a) Proceed towards 10 NM East/West of GGT (as appropriate) descending to 1000 FT.

b) Descend to circuit altitude and join traffic circuit as follows:

| **Assigned Runway** | **Approach Direction** | |
|---|---|---|
| | **East of Runway** | **West of Runway** |
| **Runway 02** | **Right Hand Downwind** | **Left Hand Downwind** |

| Runway 20 | Left Hand Downwind | Right Hand Downwind |
|---|---|---|

c) Orbit at downwind position and look out for traffic and visual signal from ATC.
d) Land on Runway and vacate to dispersal. Military Helicopter may land on Helipad on Airforce dispersal.

**VEGT AD 2.23 ADDITIONAL INFORMATION**

1.BTN 1230-0030 UTC next day. Daily Solar power obstacle light (SPOL) functioning on:

i.Sajanpara Hill BRG 142.5 DEG, HGT 755FT, DIST 2.3 NM from ARP.

ii.Barashilla Hill BRG 101.5 DEG, HGT 712 FT, DIST 2.5 NM from ARP.

iii.Obstacle light provided on Mirza Hill BRG 247.5 DEG, HGT 748FT, DIST 2.54 NM from ARP with main Power supply. SDBY Power supply not AVBL.

2.Ambulift facility available. Twelve (12) hours prior notice is required to avail the facility.

3.High thrust may affect building in the operational area. Pilots to taxi on low Power.

4.ADS-B Ground Equipment Commissioned and Operational

5.Taxiway 'E' and Taxiway 'G' declared as hot spot. Pilots to exercise caution while taxing

**6. Low Visibility Procedure Guwahati Airport (VEGT):**

**6.1 BACKGROUND:**

Until the latest amendment of DGCA Civil Aviation Requirements (CAR) Section 8, Series 'C', Part-I on All-weather Operations, Low Visibility Procedures were required at aerodromes for the purpose of ensuring safe operations during Categories II and III approaches and/or Low Visibility Take-offs (LVTO).

However, in latest amendment to CAR (Rev. 10) Para 5.3 following provision regarding Low Visibility Procedures is added. "Quote" **An operator shall not conduct take-off with RVR/visibility less than standard category I conditions of 550M RVR/800M visibility unless Low Visibility Procedures are enforced.**' "Unquote"

This provision necessitated the need of Low Visibility Procedures for accommodating/permitting departures in visibility/RVR less than 800M/550M even at such airports where there are no CAT-II and CAT-III operations.

Accordingly, Low Visibility Procedures have been developed for Guwahati Airport to accommodate/permit departures in visibility/ RVR less than 800M/550M from RWY 02 (Runway Served With RVR Instruments).

**6.2. DEFINITIONS:**

**6.2.1 Low Visibility Procedures (LVP):** Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

**Note**: as per para 5.3 of CAR on All Weather Operations, an operator shall not conduct Take-off with RVR/Visibility less than standard CAT-I conditions of 550m RVR/800m Visibility unless low visibility procedures are enforced.

**6.2.2 Manoeuvring Area:** That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

**6.2.3 Runway Visual Range:** The range over which the pilot of an aircraft on the centerline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline.

**6.2.4 Aerodrome Operating Minima:** The limits of usability of an aerodrome for:

a. Take off, expressed in terms of runway visual range and / or visibility and, if necessary, cloud conditions.

b. Landing in 2 D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and

c. Landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height(DA/H) appropriate to the type and/or category of the operation.

**6.2.5 Visibility** - Visibility for aeronautical purposes is the greater of:

a. The greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;

b. The greatest distance at which lights in the vicinity of 1000 candelas can be seen and identified against an unlit background.

**Note 1.** — The two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).

**Note 2.** — The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

**6.3. GENERAL:**

The Low Visibility Procedure (LVP) incorporates safeguarding measures to mitigate runway incursions and defines operational restrictions to ensure safe Airside Operations taking into account the available Aerodrome facilities.

**6.4. MINIMUM REQUIREMENTS:**

The following aeronautical Ground lights and RVR equipment shall be serviceable to the required standard to support Low Visibility Procedures.

a. Runway edge lights,

b. Runway end lights,.

c. Real time TDZ RVR

d. Stand by Power supply to maintain switch over time of 1 Second for Runway Edge Lights and Runway End Lights. This requirement can be met with the help of DG Set and/or UPS.

**6.4.1 Unserviceability of Aeronautical Ground Lights/ Equipment before Implementation of LVP.**

Low Visibility Procedures will not be implemented when any of the light/equipment mentioned in para 4 above is un-serviceable or is not maintained as per the required standard.

**6.4.2 Unserviceability of Aeronautical Ground Lights/ Equipment after Implementation of LVP**

When any of the light/equipment mentioned in para 4 above becomes un-serviceable or fails to meet the required standard during periods of LVP, TWR/SMC shall advise the aircraft accordingly and LVP shall be suspended and information to this effect shall be included in ATIS broadcast.

**6.5. SAFEGUARDING PROCEDURES:**

Safeguarding Procedures (SP) are instructions for relevant airport, departments and airside operators to prepare ground services and facilities for low visibility operations in order that when LVP are implemented all Safeguarding procedures are complete.

Duty Officer Tower/Tower Supervisor will initiate and co-coordinate with all the concerned agencies for completion of safeguarding procedures before implementation of Low Visibility Procedures.

**6.5.1 Safeguarding Procedures shall be initiated when:**

a. The Visibility/RVR is less than 1200M and visibility/RVR is forecast to
deteriorate to 800M or less; and/or
b. The cloud ceiling is less than 400ft and forecast to fall to 200FT or less.

**6.5.2 Safeguarding procedures include:**

a. Positioning of 1 CFT each at the two predetermined positions L2 and at fire approach road on fire access road.
b. Stopping of all maintenance works on the manoeuvring area as well as removal of all men and mobile equipment from the said area.
c. Ensuring availability of secondary power supply for change over time of maximum one second for RWY Edge and Rwy End lights supported by UPS.
(**Note:** RWY Edge and Rwy End lights may continue to operate on main power supply during safe Guarding Procedures. Whenever, LVP is to be implemented as per para 6 below, the RWY Edge and Rwy End lights shall be put on Standby Power Supply (DG set or UPS). This operation need to be completed before LVP is implemented.

*In case of DG set, Main power supply shall act as stand by power. At airports where UPS is capable of maintaining the required AGL system with one second of Switch Over time with Main Supply, the main supply can continue to be primary supply and the Generator Supply can be kept as Stand by Power supply.*

d. The appropriate Aeronautical ground lights must have been inspected during the hour preceding implementation of LVP, and thereafter once every two-hour period. These lighting inspections should be accorded priority and, if necessary, aircraft operations may be delayed.

## 6.6. LOW VISIBILITY PROCEDURES:

### 6.6.1 Implementation of Low Visibility Procedures:

Duty MET Officer shall inform Tower Supervisor whenever Visibility/RVR reduces to 800 Meters or below and/ or cloud ceiling is at 200 ft or below. Tower Supervisor shall coordinate with all the agencies to confirm whether the Safeguarding procedures have been completed or not. When Visibility/RVR falls below 800m/550M and or Cloud Ceiling is 200 FT or below and safeguarding procedures are complete, Tower supervisor will implement Low Visibility Procedures. Duty Officer Tower/Tower Supervisor shall inform all users of the imposition of low visibility procedures.

### 6.6.2 Action by various units during LVP:

a. Duty MET Officer shall keep Duty Officer Tower/Tower Supervisor informed of any change in Visibility/ RVR.

b. Apron Control shall ensure that the towing of aircraft is done under escort of “Follow Me” vehicles. “Follow Me” shall follow the route cleared by ATC;

c. SMC shall not permit any ground run on the manoeuvring area except idle power run on the stands;

d. Apron control shall ensure that “Follow Me” services are provided to pilots on request;

e. The number of the vehicles on the manoeuvring area shall be restricted to bare minimum and records of all vehicles operating on the manoeuvring area shall be maintained by SMC/Apron control.

f. The following may be included in ATIS. “Low Visibility Procedures in force”.

g. TWR shall permit departures only from the beginning of the Runway in use.

h. Whenever visibility/ RVR is less than 800/550M, Duty Officer Tower shall confirm from Pilot that the reported RVR value is within minima before issuing Take-off clearance.

i. In-Charge Electrical shall continuously monitor the main and Standby Power supply to ensure change over time of maximum one second for RWY Edge and RWY End lights during low visibility operations and report any unserviceability to Tower immediately.

## 6.7. TERMINATION OF LOW VISIBILITY PROCEDURES.

a. When Visibility/RVR improves to 800M or more and cloud ceiling is 200 feet or Higher and trend is for improvement, Tower Supervisor/Duty Officer Tower would terminate operations of LVP. He may obtain advice from Duty Met. Officer regarding improvement in weather conditions before the termination of LVP.

b. The Tower Supervisor will intimate Apron Control/SMC/ARFF/ In-charge electrical-Engineering regarding the termination of LVP operations.

c. On cancelling of LVP, following message shall be included in two subsequent ATIS broadcasts. “**LOW VISIBILITY PROCEDURES CANCELLED**”.

d. If SP are implemented and LVP are not subsequently implemented and the visibility/RVR improves and is more than 1200m and/or the cloud ceiling is 400ft or higher and both are forecast to remain above the required SP criteria Tower Supervisor/

Duty Officer Tower may cancel SP.

**6.8 ACTIONS BY OTHER AGENCIES (AIRLINES, REFUELING COMPANIES, CATERING AGENCIES, ETC.)**

a. Every Year before commencement of monsoon/winter season, a meeting will be held by Airport Director, to inform all airlines and agencies operating at airport about their roles/responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.

b. All the agencies shall ensure that staff and drivers are suitably trained during Low Visibility operations.

c. A refresher program for ATCO's and personnel responsible for airside operations shall be conducted every year.

d. All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations operate in the operational area during periods of low visibility. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.

e. All the vehicles must have their obstruction lights "ON" during Low Visibility Procedures operations.

f. All instructions/sign boards provided for vehicular movement area/service roads, must be followed while operating in the operational area.

7. ASMGCS commissioned to enhance visual surveillance.

8. Bird activty observed in and around the airfield. Pilots to exercise caution while landing and takeoff.

9. **Exemption Granted by DGCA:**

DGCA India has granted exemption vide letter dated 12.09.2022 for non-compliant width of runway strip against provisions in CAR Section 4, Series B, Part I, Para 3.4.3 for the period from up to 31.12.2025.

10. AD minima for take off is RVR 550M and ABV DUE RWY GUARD LGT not available.

11. Y2 Pushback PSN not AVBL for OPS while ACFT is TAX via TWY A to APN 1.Pilots to EXER CTN.

12 ACFT SUITABILITY OF TAXILANE D3 IS CODE C.)

13.The T2 Apron is made available with Multiple Aircraft Receiving Stand (MARS) 21L, 21, 21R, and Aircraft Stands 22L, 22R, 23L, 27 and 28. The details of facilities are as follows:

| STAND NO. | COORDINATES FOR AIRCRAFT STAND | SUITABILITY | PCR | ELEV. (FT) (EGM 08) | REMARKS | CONTACT/ REMOTE |
|---|---|---|---|---|---|---|
| **STAND 21 L** | | | | | | |
| **Stop Position A** | **260722.91N 0913553.39E** | **A21N** | **1420/R/D/ W/T** | **161.30** | **Power-in Pushback** | **Contact** |
| **Stop Position B** | **260722.96N 0913553.25E** | **B737, B739, B38M, A320** | | | | |
| **Stop Position C** | **260722.98N 0913553.19E** | **B734, B737, B738, B739, B38M, A318, A319** | | | | |
| **STAND 21** | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Stop Position A** | **260721.9N 0913553.45E** | **B77W, A346** | **1420/R/D/ W/T** | **161.30** | **Power-in Pushback** | **Contact** |
| **Stop Position B** | **260721.95N 0913553.31E** | **B789,B753,B77L A332, A333** | | | | |
| **STAND 21R** | | | | | | |
| **Stop Position A** | **260721.55N 0913553.21E** | **A21N** | **1420/R/D/ W/T** | **161.30** | **Power-in Pushback** | **Contact** |
| **Stop Position B** | **260721.61N 0913553.06E** | **B737, B739, B38M, A322** | | | | |
| **Stop Position C** | **260721.63N 0913553.01E** | **B734, B739, B738, B38M, B737, A318, A319** | | | | |
| **STAND 22L** | | | | | | |
| **Stop Position A** | **260720.33N 0913552.2E** | **A32N** | **1420/R/D/ W/T** | **161.30** | **Power-in Pushback** | **Contact** |
| **Stop Position B** | **260720.39N 0913552.06E** | **B737, B739, B38M, A322** | | | | |
| **Stop Position C** | **260720.41N 0913552.0E** | **B734, B737, B738, B739, B38M, A318, A319** | | | | |
| **STAND 22R** | | | | | | |
| **Stop Position A** | **260718.98N 0913552.01E** | **A21N** | **1420/R/D/ W/T** | **161.30** | **Power-in Pushback** | **Contact** |
| **Stop Position B** | **260719.03N 0913551.86E** | **B737, B739, B38M, A322** | | | | |
| **Stop Position C** | **260719.05N 0913551.81E** | **B734, B737, B738, B739, B38M, A318, A319** | | | | |
| **STAND 23L** | | | | | | |
| **Stop Position A** | **260717.75N 0913551.01E** | **A21N** | **1420/R/D/ W/T** | **161.30** | **Power-in Pushback** | **Contact** |
| **Stop Position B** | **260717.81N 0913550.86E** | **B737, B739, B38M, A320** | | | | |
| **Stop Position C** | **260717.83N 0913550.81E** | **B734, B737, B738, B739, B38M, A318, A319** | | | | |
| **STAND 27** | | | | | | |
| **Stop Position A** | **260720.83N 0913544.88E** | **AT72** | **690/R/D/ W/T** | **161** | **Power-in Pushback** | **Remote** |
| **STAND 28** | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Stop Position A** | **260719.79N 0913543.84E** | **A21N** | **690/R/D/ W/T** | **161** | **Power-in Pushback** | **Remote** |
| **Stop Position B** | **260719.79N 0913543.88E** | **B737, B739, B38M, A322** | | | | |
| **Stop Position C** | **260719.74N 0913543.99E** | **B734, B737, B738, B739, B38M, A318, A319** | | | | |
| **Stop Position D** | **260719.71N 0913544.07E** | **A318, DH8D** | | | | |
| **Stop Position E** | **260719.69N 0913544.12E** | **AT72** | | | | |

## VEGT AD 2.24 CHARTS RELATED TO AN AERODROME

1 Aerodrome Chart (13 June 2024)
2 Aircraft Parking / Docking Chart APRON I (19 March 2026)
3 Aircraft Parking/Docking Chart Apron K (19 March 2026)
4 Aerodrome Obstacle Chart Type-A (Operating Limitations) RWY 02 (22 January 2026)
5 Aerodrome Obstacle Chart Type-A (Operating Limitations) RWY 20 (22 January 2026)
6 Aerodrome Obstacle Chart - Type B (22 January 2026)
7 Aerodrome Obstacle Chart - Type B - Aerodrome Obstacle Tabulation 1 (22 January 2026)
8 Aerodrome Obstacle Chart - Type B - Aerodrome Obstacle Tabulation 2 (22 January 2026)
9 Aerodrome Obstacle Chart - Type B - Aerodrome Obstacle Tabulation 3 (22 January 2026)
10 Aerodrome Obstacle Chart - Type B - Aerodrome Obstacle Tabulation 4 (22 January 2026)
11 Instrument Approach Chart - VOR Procedure RWY 02 (17 April 2025)
12 Instrument Approach Chart - VOR Procedure RWY 20 (16 May 2024)
13 ATC Surveillance Minimum Altitude Chart (16 May 2024)
14 Instrument Approach Chart - RNP Procedure RWY 02 (16 May 2024)
15 Tabular Description - RNP Procedure RWY 02 (16 May 2024)
16 Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 02
17 Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 02-Coding Table
18 Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 02-Waypoints
19 Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 20
20 Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 20-Coding Table
21 Standard Departure Chart – Instrument, RNAV1 (GNSS), RWY 20-Waypoints
22 Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 02
23 Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 02-Coding Table
24 Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 02-Waypoints
25 Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 20
26 Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 20-Coding Table
27 Standard Arrival Chart – Instrument, RNAV1 (GNSS), RWY 20-Waypoints
28 Instrument Approach Chart - ILS or LOC Procedure RWY 02 (17 April 2025)
29 Aircraft Parking/Docking Chart – Apron T2 (19 March 2026)
30 Aircraft Pushback Procedure Chart – Stand 21L, 21R, 22L, 22R & 23L (19 March 2026)
31 Aircraft Pushback Procedure Chart – Stand 21 (19 March 2026)
32 Aircraft Pushback Procedure Chart – Stand 27 & 28 (19 March 2026)

## VEGT AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

| Procedure | Procedure Minima | VSS Penetration |
|---|---|---|
| **1** | **2** | **3** |
| | | |